

Predicting Molecular Trajectories of a Nanomotor with Neural Networks

WHK Research Project

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Presentation outline

1. Scope of my work
2. Dataset overview
3. Training the network
4. Results
5. Possible next steps

Scope of current work

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- ▶ Work on one specific dataset:
MOTOR_MD_XTB/T300_1
- ▶ Prediction of **energy** and **forces**
given atomic positions
- ▶ Molecule is only examined in
ground state

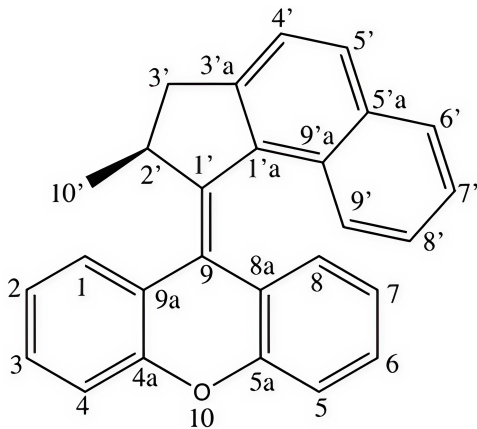
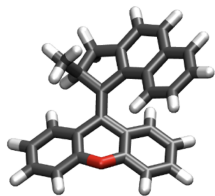


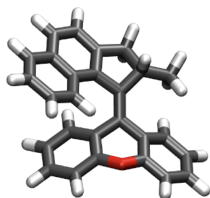
Figure: Figure adapted from [1, p.13]

Dataset overview

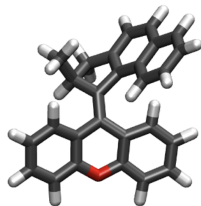
Dataset overview



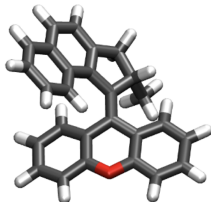
anti-M



anti-P



syn-M



syn-P

- ▶ **Four important conformations:**
anti-M, syn-M, anti-P and syn-P
- ▶ Can be roughly clustered by using **dihedral angles**, e.g.

$$\text{M: } \beta < 0, \quad \text{P: } \beta > 0$$

and furthermore

$$\begin{aligned} \text{syn-M: } \beta < 0 \quad \& \quad \delta < 0, \\ \text{anti-M: } \beta < 0 \quad \& \quad \delta > 0 \end{aligned}$$

Figure: Figure adapted from [1, p.11]

Dataset overview

- ▶ Data sampled in **NVT ensemble**
- ▶ Split in train (12615), validation (3154) and test data (3943)
- ▶ **Center of mass** and **energy offset** subtracted
- ▶ Configurations in T300_1 mostly in **anti-M**

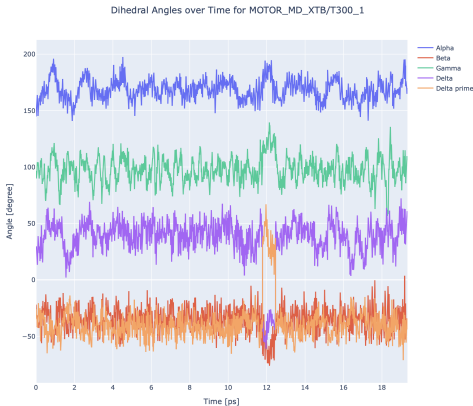


Figure: Plot of dihedral angles for T300_1 taken from [3].

Training the network

Training the network

- ▶ Training using SchNetPack (see [2] and Pragati's talk)
- ▶ Mean squared error (MSE) is underlying loss function
- ▶ Energies weighted with 0.01 and forces with 0.99
- ▶ Runs with 1000 epochs take roughly six hours and don't converge

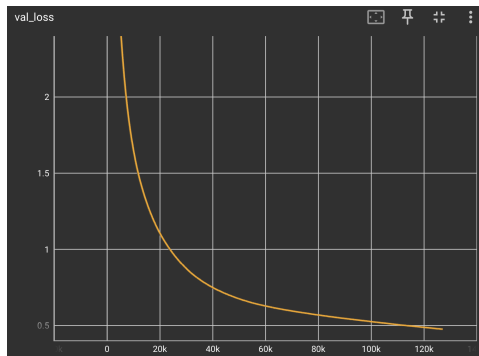


Figure: Validation loss plotted against the number of steps when training a model on Angstrom and kcal/mol for 1000 epochs.

Impact of units

- ▶ Change of units (COU) **affects model performance** (see results section)
- ▶ Possible explanation: COU **affects weights** of energies and forces within the MSE
- ▶ Trying to counteract this effect, is not yet fruitful (see starred entry in results section)
- ▶ More on this later, if you are interested!

Results

Performance on Validation Data

Molecule	Training Units	Energy MAE [kcal/mol]	Forces MAE [kcal/mol/Å]
CPNX	Å, kcal/mol	0.1896 ± 0.0134	0.2999 ± 0.0110
CPNX	Å, eV	0.1581 ± 0.0027	0.3174 ± 0.0067
CPNX	Å, Hartree	0.2960 ± 0.0050	0.5154 ± 0.0100
CPNX	Å, Hartree*	0.3053 ± 0.0085	0.4993 ± 0.0119
$C_9O_8H_4$	Å, kcal/mol	0.11 – 0.38	0.14 – 1.17

Table: Mean absolute errors (MAEs) with corresponding standard errors on the validation data, for models trained in three different unit systems. The reference values for Acetylsalicylic ($C_9O_8H_4$) are taken from [2, p.453]. Note that the chemical accuracy is more or less defined as 1 kcal/mol

XTB vs. NN Trajectory

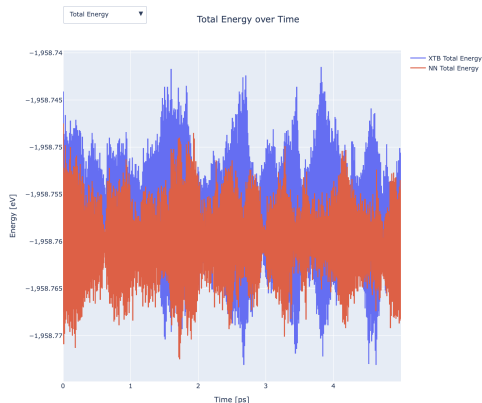


Figure: Plot of predicted energies from XTB and the neural network taken from [3].

- ▶ Take last sample of test data as starting configuration
- ▶ Using its velocities should ensure that **trajectory is equilibrated**
- ▶ Predict trajectory with XTB and neural network for 5000 steps à one femtosecond
- ▶ Simulating this with NN takes around five minutes

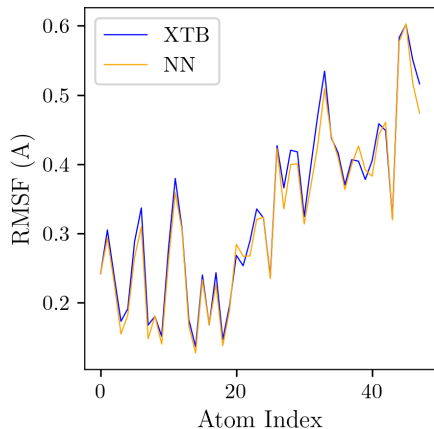
Correlation

- ▶ How similar are the trajectories around a certain point?
- ▶ Correlation as **measure of similarity**
- ▶ At step s take the next n steps to compute the **Pearson correlation**
- ▶ Repeat for each time step



Figure: Plot of correlations between predicted energies from XTB and the neural network taken from [3].

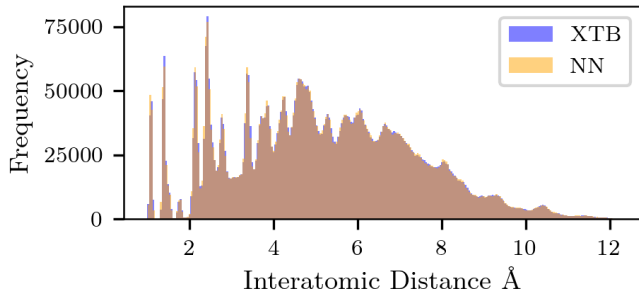
Per-atom fluctuations



- ▶ Take average over positions to compute reference structure
- ▶ Compute root mean squared fluctuations (RMSF) from reference structure per atom over time
- ▶ Indices $[0, \dots, 47] \hat{=} \text{atoms } [O, C, \dots H, \dots]$
- ▶ Pearson correlation of 0.9934

Figure: Average RMSF per atom.

Interatomic distances



- ▶ Plot of distribution of interatomic distances
- ▶ $48 \cdot 47 / 2 = 1128$ distances per step
- ▶ Flattened into array of length $1128 \cdot \text{\#steps}$

Radius of gyration

- ▶ Radius of gyration is the distance from a systems **rotational axis** to its **center of mass**
- ▶ Describes **how masses** of system are **distributed**

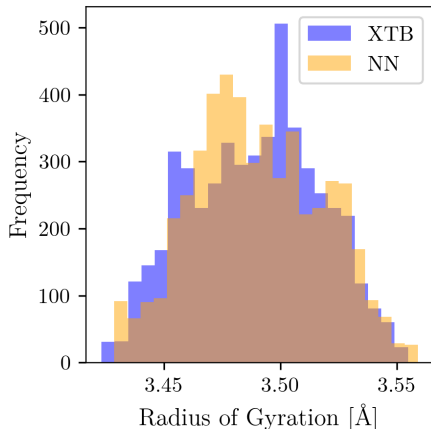
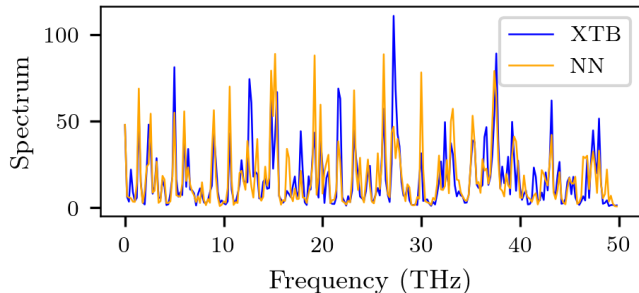


Figure: Distributions of the radius of gyration.

Power spectrum



- ▶ Excerpt of power spectra from XTB and NN trajectories, $r_{\text{Pearson}} = 0.7125$
- ▶ Calculated as Fourier transform of the velocity autocorrelation function
- ▶ Gives insights into active vibrational modes

Possible next steps

Possible next steps

- ▶ Include also the other T300 datasets in training and inference
- ▶ Train models until they converge
- ▶ Move on to excited states
- ▶ Try to use trained model as foundational model:
 - ▶ Finetune it for specific needs
 - ▶ Perform MD on the fly

Sources

- [1] Jesús Lucia-Tamudo, Michelle Menkel-Lantz, and Enrico Tapavicza. “First principles prediction of wavelength-dependent isomerization quantum yields of a second-generation molecular nanomotor”. In: *Physical Chemistry Chemical Physics* 27.23 (2025), pp. 12519–12531.
- [2] KT Schütt et al. “SchNetPack: A deep learning toolbox for atomistic systems”. In: *Journal of chemical theory and computation* 15.1 (2018), pp. 448–455.
- [3] Ferdinand Tölkes. *Molecular Dynamics of a nanomotor using SchNetPack*. <https://github.com/FerdinandToelkes/whk>. GitHub repository. 2025.